



## DPUCounter

### 1. Inputs:

|              |                                                                                  |
|--------------|----------------------------------------------------------------------------------|
| <i>Start</i> | Starts the counter if it changes from 0 to 1.                                    |
| <i>Stop</i>  | Stops the counter if it changes from 0 to 1.                                     |
| <i>Reset</i> | Resets the counter if it changes from 0 to 1.                                    |
| <i>Delta</i> | Value by which the counter is incremented in each simulation step.<br>Default: 1 |

### 2. Outputs:

|            |         |
|------------|---------|
| <i>Out</i> | Counter |
|------------|---------|

### 3. Functional principle:

Let *Out'* be the value of the counter from the previous simulation step. Then the DPU computes in each simulation step

$$Out = Out' + Delta$$

The counter is started at 0 as soon as *Start* changes from 0 to 1. It is stopped as soon as *Stop* changes from 0 to 1. If *Start* and *Stop* are set at the same time, the counter is incremented only for one simulation step.

### 4. Example:

This will count how many times the indicator has been activated since the last reset:

```
DPUCounter IndicatorCount
{
    Computer = { MOCKUP };
    Index = 20;
};
Connections = {
    Mockup.Indicator -> IndicatorCount.Start,
    Mockup.Indicator -> IndicatorCount.Stop
};
```